Lean 6 Sigma Black Belt

Pre-course Reading

Please read this thoroughly before attending the
Black Belt Course as it covers essential pre-course preparation

Lean Sigma Black Belt – pre course preparation

Firstly thank you for booking your Black Belt Lean Six Sigma course with us. We very much look forward to meeting you on the course and helping you expand the knowledge you gained on your Yellow and Green Belt courses.

This document was written to help you prepare for the Black Belt course which you will be attending shortly. As is the case with all courses, preparation is important in order for you to maximise the benefit from your investment. In the case of the Black Belt course this is doubly important as the training is a continuation of our Yellow & Green Belt courses with which we expect you to be thoroughly familiar. Reference will be made to these courses in the class and in the Black Belt exams.

If you did not study Yellow & Green Belt (Y/G) with us, or you no longer have your course manuals please contact your sales advisor as soon as you can.

What is the focus of the Black Belt course?

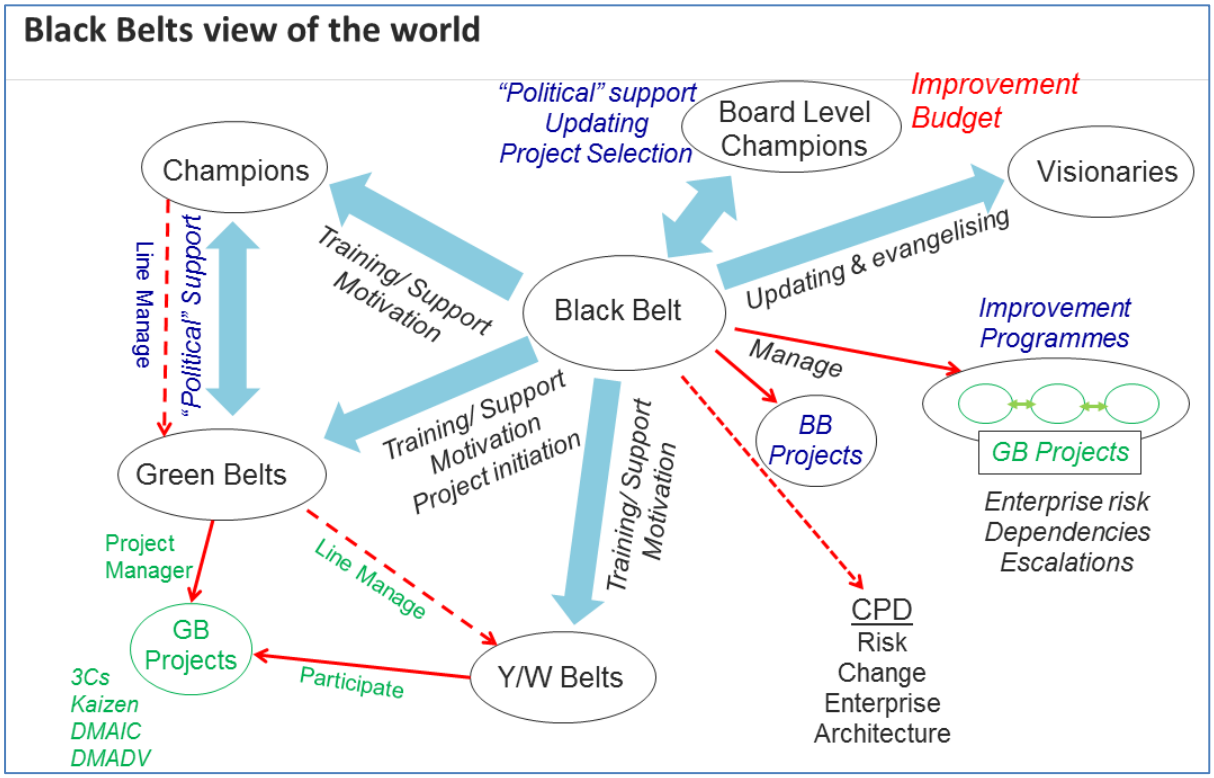
Our view of the role of the Black Belt can be summarised as follows:

Black Belt Skill Set

- Mentoring, Coaching & Leadership skills
  - Especially for Green Belts
- Technical Escalation & Training
- Ability to handle larger projects/ small programmes
- Understanding of practicalities of Statistical Modelling such as:
  - Control Chart types
  - Hypotheses Tests
  - ANOVA
- Presentation & Liaison to Senior Management
- Communication of the Lean 6 Sigma vision

A hands on approach beyond Green Belt

As you can see the Black Belt lives in the Business world as much as the technical world:



The Black Belt course is a technical course aimed at giving you the skills needed to perform statistical modelling of a process. It will also get you started on the people and communications side of the Black Belt role and develop you understanding of Lean Metrics. You will not be attending a course on statistics rather you will be attending a course on the practical application of statistical tools to help you understand how your business runs and how it could be improved

The course relies on you having a sound grasp of the Yellow & Green Belt syllabus (**you MUST be qualified to Lean 6 Sigma Green Belt level before attending the Black Belt course**) so it is important that you revise the work you performed on those courses.

Here is a heads up on the makeup of the Black Belt course:

The Course at a glance

What the course is

- The second week of the Black Belt training programme
- Builds on, and assumes familiarity with, the Yellow & Green Belt syllabus

What it is not

- A statistics course
  - We are interested in the practical application of statistics to help is improve processes
- A Minitab course
  - Minitab is the statistical package we use in our examples
- A formal Project/ Programme Management course
  - Black Belts may need skills in this area however this is not the focus of this course
- A consultancy
  - To really consult your trainer would need to be familiar with your business. We can offer advice however (but it is up to you to decide if it is relevant to your situation!)

Course Flow

This is the way the course is put together:

Day 1
Introductions
The Role of the Black Belt
Black Belt DMAIC
Physical Cell design & Optimising Flow
Black Belt as a trainer

Day 2
Statistical foundation
Process Capability
Hypotheses tests
Design of Experiments (DoE)

Day 3
Probability distributions
Advanced Control charts

Day 4
Measurement System Analysis (MSA)
Lessons from Queue Theory
Lean Metrics

Day 5
Catch up & loose ends
Final Task
Black Belt Exam

Evenings
Revision of course
Work and index the Body of Knowledge (BoK)
Exercises

Don't forget there are exercises and discussions.

Black Belt Assessment

Your understanding of the Black Belt syllabus (which includes the Yellow & Green Belt syllabus) is tested by class assessment (50%) and a final exam (50%). You will need a combined score of 70% to be awarded a pass. Minitab & evening work feature heavily in the assessments.

What will I need to bring with me to the course?

You will need to bring the following to the course:

Minitab 16 or above *Minitab is a sophisticated statistical program which is popular with Lean 6 Sigma professionals* *You need this to perform class exercises* *You will also need the ability to load data files onto your computer – our preferred method is the trusty USB stick. If this is not suitable for you please contact us before the course so we can find a solution for you*	You will need this to perform the exercises in the class and in the homework Later sections of this guide provide advice on obtaining & installing Minitab **WARNING:** *The 30 day trial can be installed on a computer only* ***once*** *so make sure the trial does not expire before the end of the course* *Many of the exercises and activities require data files. These will be provided by your Instructor during the course. The data files are formatted for Minitab version 17 earlier versions may not work with them.*
Calculator	Handy to perform Lean Metric calculations in the class, in the homework, and in the exam You do not need a scientific calculator. A simple cheap model from a chain store is sufficient. However make sure it can perform square roots
Your Yellow & Green Belt course manual *This is the course workbook issued on the Yellow & Green Belt courses*	The Black Belt course builds on the Yellow & Green Belt course and the knowledge contained in this manual is integral to your understanding. The exam can test on the full syllabus which includes the Y/G belt courses from version 8 onwards. Later sections of this guide offer advice on what to do if you did not study Green Belt with us or if you have lost your manual
Stationery	Paper, writing tools, highlighter and page tabs

Note: *You must have Minitab installed onto your laptop before arriving on the course. We cannot offer technical support in this area for many reasons. Also we do not keep spare copies of Yellow & Green Belt courseware at our locations (they are usually printed on demand).*

How should I prepare for the course?

It is essential to prepare for this course. Many people leave a gap between their Green Belt and their Black Belt training. This has advantages but unfortunately you will find that some things are forgotten – so I'm afraid you are going to have to "hit the books"

To refresh your Green Belt Knowledge,

Read through your Y/G Belt course notes to get yourself back up to speed – use the following lists as a guide:

Lean Topics assumed on the BB Course

DMAIC – the activities and typical tools used in each phase

3Cs

Kaizen

Basic Lean Metrics (Takt time, Cycle time)

VOC & SIPOC

Data collection planning

Seven wastes of Lean

Poka Yoke

5S

SMED

Value Stream Maps

Cell Management

Basic Risk Management

Process Levelling & Flow control

Facilitated Workshops & consensus building tools (Pugh matrix etc)

Basic small-scale project management

Statistical Topics assumed on the BB Course

The concept of SPC

Means

Standard Deviation

Basic Control Charts including Control & Specification Limits

Common & Special cause variation and Control Chart Special Cause identification rules

Process Sigma, Yield & DPMO

Basic Process Capability

Data types (Variables/Discrete/Binary)

Assemble your tools

Minitab is the major tool you will need on the course – it is essential to have this statistical programme up and running before you attend the course. Details are given later in these notes.

Also assemble the stationery of your choice writing materials, tabs, and a simple calculator.

Sort out your diary – there will be study in the evening!

Read the Scenario and perform the task

As part of your Black Belt assessment you will be asked to perform a number of common Black Belt tasks. Some of the tasks will be based on the class scenario (Uncle Jim's for those of you who remember our Green Belt course). It is similar to the GB Case Study but it is not quite the same and it is not necessary for you to be familiar with it. Your first task is to perform a small presentation as outlined later in this document.

An excerpt from the scenario and the task description is included at the end of this document

One last point:

The Black Belt exam is a paper based exam consisting of multiple choice questions. It tests you on the information contained in the Body Of Knowledge (BOK) as defined in our Yellow, Green and Black Belt course manuals.

If this type of exam is a problem for you then it is critical that you inform us well in advance so that we can discuss how we can support you.

What to do about missing manuals

As we may have hinted, you need the knowledge contained in the Yellow & Green belt manuals. Our Black Belt course is designed to continue from the our Green Belt (which in turn depends on our Yellow Belt) and there is not enough time in the five days of the Black Belt course to go over these topics again.

You need to have a recent copy (at least version 8) of our workbook so if you have an earlier version, or did not receive training in Green Belt from us, then please contact your sales advisor straight away and ask them to arrange for a new set to be shipped to the course venue.

What to do about Minitab

Minitab is the best known and most common statistical package used by 6 Sigma practitioners. It is a Windows based application, it does not work on Apple macs so please ensure you bring a windows computer with Minitab ready to go.

We feel it is a key skill for Black Belts to be able to demonstrate the ability to use a statistical package when modelling processes. There are other statistical software packages available, however we believe Minitab has developed its capabilities to match the evolution of Lean Six Sigma. It also is the 'most' popular statistical software so there is a good chance that your customers or supplier will be using it. All the statistical exercises, the course graphics and the exams are based on this product.

You can download and install a 30 day trial version of Minitab onto most Windows Based PCs: Google Minitab or go to Minitab.Com

Tip: *These are just some of the problems people on our courses have had when installing the evaluation copy of Minitab.*

The evaluation install failed as it had previously been installed

The evaluation timed out as it was installed prior to 30 days

Installed in wrong language

Locked down corporate computer blocked the install

It doesn't install on my Mac!

It doesn't work in Open Office

This is why it is important to have Minitab up and running in advance of the course – we can't do this for you I'm afraid

Uncle Jim's Sweet Factory

Introduction to the Black Belt Course Scenario

This section gives you an overview of the Case Study used on the Black Belt course – the aim is to introduce you to the background which is used in several of the exercises and allow you to prepare for the first exercise

Note: *This is an adaptation of our Case Study suitable for this stage of your preparation. Do make sure you review the full work during the course*

Background

Consolidated Sugar & Fats (CS&F) is a sweet manufacturer based in the small market town of Minton Blizzard. Although the company was originally privately owned it has been a limited company since the mid-1990s. The company's main brand is "Uncle Jim's" which has a worldwide appeal partly due to its nostalgic associations but also because of the excellent quality of the sweets.

In this case study we are going to limit ourselves to a particular type of confectionary known as "boiled sweets".

CS&F have 3 brands of boiled sweet which you need to know about:

"Uncle Jims" Brand

The original product upon which the company was founded. First made by hand in the 1920s the brand has the biggest profit margin, is unapologetically nostalgic, and sells world-wide.

"Dibbers Supermarket"

CS&F have a contractual obligation to provide Dibbers with their "Own Brand" boiled sweets.

"Target" Brand

Devised by CS&F as a way to help mitigate their fixed costs. When there is no need to produce the other brands then the production line makes Target. Value sweets, priced to sell, CS&F do not guarantee supply of these sweets.

A short overview of the boiled sweet manufacturing process

CS&F have two production lines that manufacture boiled sweets. The original process was semi-automated in the late 1970s and has been maintained ever since. This process is crude by modern standards but does give an "old fashioned" feel to the sweets made due to the variance of the sweet forming process. The line consists of a number of independent units. Some have been replaced with more modern models over the years but the sweet formation, which gives the sweets their old-fashioned look and feel, remains unchanged.

In your talks with the company you have discovered that they appear to be concerned especially about the quality of the "wrapping station" (the unit that wraps each sweet in a plastic film). Uncle

Jim's uses only natural ingredients their confectionery, each sweet consists mainly of sugar with a small percentage of flavourings added.

The sugar and other ingredients are melted in a boiler unit (boiler) which is positioned over a moving belt. The boilers are sealed units similar to pressure cookers. As the mixture heats up it melts and the internal pressure builds up. The pressure forces the viscous mixture through a nozzle which has a number of holes. The pressure forces the molten mixture through the holes in discrete drips. The falling drips impact the belt which results in the characteristic "boiled sweet" shape.

The sweets then flow under a cooling fan and then pass into an automatic wrapping device which wraps them in a plastic film.

The wrapped sweets are collected into a hopper which has a funnel at the bottom with an automatic dispensing valve.

Opened bags travel under the valve which feeds a pre-set weight (nominally 100) of sweets into each bag. The bags then travel to a sealing station which closes the bags. The bags are then auto packed into cartons of 60 bags and the cartons are sealed ready for shipping.

The factory has two production lines; this means that the company can produce two types of sweet at a time. To manufacture another type of sweet the factory needs to stop a production line and thoroughly clean it to remove any trace of the previous ingredients - this takes about 2 hours so company policy is to manufacture in large batches (typically they produce enough of one sweet to satisfy demand for 3 months).

Major Business Units

CS&F has the following major units in the business:

Unit	Value add
Purchasing	Negotiates and orders ingredients for sweet creation (sugar and flavourings)
Goods inward	Stores supplies when they arrive transfers them to Manufacturing on a weekly basis
Manufacturing	Produces the actual sweets. Bags and boxes them
Goods outbound	Picks up the finished boxes from manufacturing, stores them (in the Dispatch Warehouse) and hands them out to authorised collectors (when released by sales)
Sales	Sells both Target and Uncle Jim's brands Has a dedicated team and Account Manager for Dibbers Has a dedicated Business Unit for Web fulfilment
Web fulfilment	Handles sales orders and customer queries generated from the UncleJimsSweets.co.uk web site. This is a growing sales channel although not properly integrated into the rest of the company.
Marketing	Responsible for branding, package design, new products. Also handles PR
Compliance	Ensures that products meet International, National, and Corporate standards. Contains the Chemical Analysis team who check materials and products Performs internal Product Quality Assurance
R&D	Research and development acts as an "Internal Consultancy" to the rest of the company. This is a small department of scientists and their support team. R&D provide other departments with specialist advice as needed.
H&S	Health & Safety Ensures that employees work in a safe and healthy environment. Particularly interested in the ingredients used in production and the details of how such materials can be safely managed. Responsible for creating and maintaining OTGIs (On The Ground Instructions) given to staff concerning materials handling and storage
FM	Facilities Maintenance Cleaning and maintenance of infrastructure The FM Manager is also responsible for the maintenance of the manufacturing machinery (including fork lift trucks etc). This is provided by a mixture of dedicated CS&F employees and service contracts
IT	Handles IT systems, Help Desk, installation, upgrades and support. Controls ERP system and admin packages
Corporate Services	HR, Finance, Admin, Canteen Supervision (Canteen is self-financing but employs full time or part time staff

Assessment task 1: Elevator pitch (2 marks)

Imagine that you have been introduced to Alan Forden the MD (CEO) of CS&F. Alan is very keen to take up Lean 6 Sigma but is having difficulty explaining to his fellow Board members why it would be a good idea. Evidently, some members of the Board have expressed scepticism as they don't think that quality initiatives are capable of driving costs down in a significant way.

Alan has asked you to attend the next board meeting and provide a very short overview of what Lean 6 Sigma is and why CS&F should embrace Lean Thinking.

What you need to do:

Prepare a **short** (3 minutes maximum!) presentation explaining how embracing Lean 6 Sigma will be beneficial to CS&F and respond to the Board members concerns.

As an example of how simple and effective Lean Thinking can be, identify two simple tools which you could teach to CS&F employees that would help them develop a Lean viewpoint straight away (use tools discussed in the Yellow or Green Belt courses). Describe these in your presentation.

You will be given a chance to give this presentation during the first day of the course. Successful completion of this task will count towards your Black Belt assessment

Tip*: Keep this short and sweet – you only have 3 minutes to deliver your presentation. Black Belts need to be communicators and good communicators know their audience and adapt their presentations to fit. You audience is the Board of Directors of CS&F and the aim is to grab their attention and make them think that Lean thinking is a good philosophy to invest in.*